

IPv4, IPv6, NAT, and ICMP and Transport Layer, UDP, and Reliable Data Transfer

This report explores the critical architectural bridge between the Network Layer (handling host-to-host delivery) and the Transport Layer (facilitating process-to-process communication), as detailed in Lectures 17 and 18

1. **Inside the Network Layer:** Lecture 17 shifts from the "how" of routing to the "what" of the packet, examining the machinery that makes global connectivity possible

The IPv4 Datagram and ICMP Diagnostics

Every IPv4 packet begins with a minimum 20-byte header containing fields essential for router processing.

- Time to Live (TTL): This field acts as a "hop counter," decremented by each router to prevent packets from looping infinitely.
- Protocol Field: This 8-bit key (e.g., 1 for ICMP, 6 for TCP, 17 for UDP) tells the destination host which transport-layer module should receive the payload.
- ICMP (Internet Control Message Protocol): Because IP is unreliable and connectionless, ICMP provides a feedback loop for error reporting. It powers diagnostic tools like ping (using Echo Request/Reply) and traceroute, which discover network paths by intentionally triggering "TTL Exceeded" error messages from routers.

Fragmentation and the NAT "Hack"

The sources highlight two major challenges in IPv4: link MTU mismatches and address exhaustion.

- Fragmentation: When a packet exceeds a link's Maximum Transmission Unit (MTU), routers must split it into fragments, which are reassembled only at the destination. This is complex and wasteful, leading modern systems to use Path MTU Discovery (PMTUD), which uses ICMP to find the smallest MTU along a path to avoid fragmentation entirely.
- NAT (Network Address Translation): This mechanism allowed private networks to share a single public IP address, effectively "saving" the internet from IPv4 exhaustion. However, NAT fundamentally violates the end-to-end principle, breaking incoming connections and complicating peer-to-peer applications, which require workarounds like port forwarding or UPnP.

The IPv6 Vision and DHCP

IPv6 was designed to replace IPv4 with a 128-bit address space, providing a virtually infinite supply of addresses (3.4×10^{38})

- Header Simplification: IPv6 removed the header checksum to speed up router processing and mandated that fragmentation only occur at endpoints.
- Structured Addressing: A key architectural rule in IPv6 is the mandatory /64 subnet size, which is required for Stateless Address Auto-Configuration (SLAAC) to function.
- Bootstrapping: For hosts to get online, DHCP automates configuration through the four-step "DORA" dance: Discover, Offer, Request, and Ack

1. The Transport Layer & The Quest for Reliability : Lecture 18 transitions from delivering packets to machines to delivering data to specific **processes**.

Multiplexing and UDP

The transport layer uses ports and sockets to demultiplex data. A socket is uniquely identified by a 5-tuple: (Protocol, Local IP, Local Port, Remote IP, Remote Port).

- UDP (User Datagram Protocol): Described as a "minimalist" protocol, UDP adds only port-based demultiplexing and an optional checksum.
- Why UDP? It is the preferred choice for low-latency applications like VoIP and online gaming, where a dropped frame is better than a delayed one, and for simple query-response protocols like DNS. Its checksum includes a "pseudo-header" with IP addresses to guard against misdelivered packets

Building Reliable Data Transfer (RDT)

To provide reliability over an unreliable network, protocols use four building blocks: Checksums, Acknowledgments (ACKs), Sequence Numbers, and Timers.

- Stop-and-Wait: The simplest RDT protocol sends one packet and waits for an ACK. It requires at least a 1-bit sequence number to allow the receiver to distinguish between a new packet and a retransmission. However, it is highly inefficient on high-speed or long-distance links (high Bandwidth-Delay Product), where the link may sit idle more than 99.99% of the time.

Pipelining: GBN vs. Selective Repeat

To fill the "pipe," protocols allow multiple unacknowledged packets to be in flight using a **window**.

- Go-Back-N (GBN): Uses cumulative ACKs (ACK n means everything up to n arrived). If a packet is lost, the sender retransmits the entire window, which is simple for the receiver but wasteful under loss.
- Selective Repeat (SR): The receiver buffers out-of-order packets and sends individual ACKs for every packet. The sender retransmits only the lost packet, making it much more efficient but requiring a more complex receiver

TCP: The Hybrid Solution Modern TCP is a sophisticated hybrid. It typically uses cumulative ACKs (like GBN) but requires a receiver buffer and often uses the Selective Acknowledgement (SACK) option to retransmit only missing segments (like SR). This design allows it to perform reliably across a vast range of network conditions